

VIKRAM VARKOOR

FPGA / Embedded Systems Engineer & Applied AI Developer

[Email](#) | [GitHub](#) | [Portfolio](#) | [LinkedIn](#)

PROFILE

Electrical Engineering graduate operating at the intersection of hardware systems and applied AI, from FPGA acceleration and embedded firmware to full-stack LLM-powered applications. Published IEEE researcher (two papers, one under peer review) with hands-on experience in IoT device validation and production AI system deployment. Comfortable across the full stack, from PCB design and RTL implementation to REST APIs and web deployment.

EDUCATION

MSc Artificial Intelligence, Heriot-Watt University, Dubai

Starting Sep 2026

Recipient of the Duffin Family Future Shapers Scholarship.

BSc Electrical Engineering, American University in Dubai

Graduated May 2026

Relevant coursework: Communication Systems, Digital Signal Processing, Systems & Controls, Microprocessors, Embedded Systems, Circuit Analysis, Electromagnetics.

EXPERIENCE

Hardware R&D Intern, Syncrow IoT, Dubai, UAE

May 2025 – Sep 2025

- Conducted structured hardware validation and functional testing of 15+ IoT devices, including Hikvision intercoms, radar presence sensors, smart locks, smart ACs, energy clamps, and multi-mode gateways.
- Designed a standardized 17-step IoT device testing procedure covering power-up, protocol handshake, network connectivity, latency, backend validation, and stress testing.
- Built a structured device catalogue documenting specs, measurement accuracy, and reliability ratings for 30+ devices.
- Authored validation requirements documentation, working cross-functionally with the product owner and data scientist.
- Worked with the Tuya platform for smart device integration; researched MQTT, ZigBee, and wired/wireless IoT architectures.

PROJECTS

Synapse.PL: HEAD System, Senior Design Project

AUD EECE 498/499

Vitis HLS · Vivado · Xilinx Zynq-7020 · Python · XGBoost · Next.js · TypeScript

- Built a Hardware-Accelerated Explainable Alzheimer's Detection (HEAD) system achieving 17-second end-to-end MRI-to-diagnosis inference on a \$269 FPGA platform, vs. \$3,000–10,000 GPU-equivalent solutions.
- Sole contributor for GLCM texture feature extraction: a local block 3D Grey-Level Co-occurrence Matrix pipeline extracting 252 Haralick features per scan across 13 spatial directions.
- Implemented the FPGA accelerator in Vitis HLS, synthesized via Vivado to the Zynq-7020 fabric: 100MHz clock, 13% LUT / 22% DSP utilization.
- Three-task cascade ML classifier (XGBoost + Optuna) achieving AUC 0.903 on 3,436 ADNI scans.
- Built a full-stack clinical dashboard (Next.js/TypeScript) with a 3D MRI viewer, Grad-CAM heatmaps, and SHAP attribution.

Deriv Trading Agent, Autonomous AI Trading System

TypeScript · Next.js · Redux Toolkit · Jest/RTL · Cypress · Python · scikit-learn · Supabase

- Built an autonomous EUR/USD paper-trading system: live candle ingestion, LLM-driven BUY/SELL/HOLD decisions with confidence scoring, and demo-account execution.
- Implemented risk management at both prompt and code level (confidence thresholds, 2:1 reward-to-risk, automatic stop-loss/take-profit).

- Built a separate statistical validation layer in Python (logistic regression, RSI/SMA/momentum features), achieving 52.51% directional accuracy vs. a 46.59% naive baseline; identified and fixed a data-leakage bug that had produced a misleading 95.88% backtest result.
- 61-test Jest/RTL suite, one component built strictly test-first via TDD, and a 9-assertion Cypress E2E suite.

Lumen, Multi-Model AI Aggregator

Next.js · TypeScript · Supabase · Groq API · Vercel

- Built a multi-agent LLM aggregator with parallel worker execution across 3 models simultaneously via SSE streaming.
- Implemented a judge orchestrator agent performing Jaccard similarity-based agreement scoring and synthesized reconciliation.
- Deployed on Vercel with Supabase Postgres for auth-gated query history and shareable links.

PaperChat, RAG-Powered Document Chatbot

Next.js · FastAPI · Python · fastembed · NumPy · Groq API

- Built a full-stack RAG application for real-time PDF chat with source citations.
- Replaced PyTorch + ChromaDB with fastembed (ONNX) + custom NumPy cosine similarity to run the full pipeline under 80MB RAM on a free-tier backend, down from ~500MB.
- SSE streaming delivers responses at ~800 tokens/sec via Groq API.

Smart Meter Analytics at Scale, Distributed Data Pipeline Benchmark

Python · PySpark · pandas · Parquet · NumPy

- Built a synthetic IoT dataset generator producing 10.5M readings across 300 devices, partitioned as Parquet for partition pruning.
- Benchmarked PySpark against pandas honestly across scale, including where Spark loses to shuffle overhead, and documented the full lazy-evaluation / DAG / Catalyst optimizer conceptual shift.

RESEARCH & PUBLICATIONS

- **IEEE ITEC-AP 2025:** Smart EV Charging Frameworks: Grid Integration and Demand-Side Optimization (Co-Author)
- **IEEE ICTMOD 2024:** Microprocessor and ML Integration: Embedded AI Pipeline Design for Edge Applications (Co-Author)
- **Under Peer Review:** FPGA-Accelerated Edge AI for Clinical Decision Support: Architecture, Implementation and Performance Analysis

SKILLS

Programming: Python, TypeScript, JavaScript, C++, MATLAB, SQL

AI & ML: LLM Integration (Anthropic, OpenAI, Groq, Gemini), CrewAI, LangChain, TensorFlow/Keras, scikit-learn, XGBoost, OpenCV, RAG, Prompt Engineering

Web & Full Stack: Next.js, React, Supabase, Tailwind CSS, FastAPI, Vercel, REST/SSE

Embedded & Hardware: FPGA (Vitis HLS, Vivado), Arduino, Raspberry Pi, PCB Design, I2C/SPI/UART

IoT & Protocols: MQTT, ZigBee, Tuya Platform, Device Validation & Testing

Data & Analytics: PySpark, pandas, NumPy, SQL, Tableau, FFT/DSP

Certifications: Develop Generative AI Applications: Get Started, IBM (Coursera), 100%